Project Report: Predicting Next Day Wildfire Spread in the United States

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Github Link: https://github.com/mollyjenkin13/ucb_mids_s207_WildfirePrediction

Abstract

Wildfires have become increasingly frequent and destructive, creating a need for accurate short-term prediction tools that can support emergency response and resource allocation. In this project, we developed deep learning models to predict next-day wildfire spread using the *Next Day Wildfire Spread* dataset, which contains 64 × 64 spatial grids of environmental, topographic, meteorological, and previous-day fire information. After preprocessing the data through normalization and exploratory analysis, we implemented three convolutional neural network (CNN) architectures: a baseline 1 × 1 CNN, which predicts each pixel independently; a three-layer CNN, which captures local spatial relationships through stacked convolutional layers; and a U-Net encoder-decoder CNN, which learns multi-scale spatial patterns through convolution, pooling, upsampling, and skip connections. The models were evaluated using metrics appropriate for the dataset's severe class imbalance, including area under the receiver operating characteristic (ROC) curve (AUC), precision, recall, F1 score, and intersection over union (IoU). Results showed that incorporating spatial context substantially improved wildfire prediction performance, with each successive architecture outperforming the previous one. After hyperparameter tuning, the final U-Net achieved a test AUC of 0.9365, a test F1 score of 0.3746, and a test IoU of 0.2305, demonstrating strong generalization to unseen wildfire events and highlighting the effectiveness of multi-scale spatial modeling for next-day wildfire spread prediction.

Introduction

Wildfires are becoming more frequent and destructive across the United States, causing loss of life, environmental damage, and billions of dollars in economic losses each year. Rising temperatures, prolonged drought conditions, reduced vegetation moisture, and expanding development in fire-prone regions have increased both the likelihood and severity of wildfire events. As wildfire activity intensifies, communities face greater risks to public health, infrastructure, and natural ecosystems.

In this project, we aim to use machine learning to predict where active wildfires in the United States will spread within the following 24 hours given the current fire perimeter and environmental conditions. By forecasting wildfire spread before it occurs, emergency managers and firefighting agencies can better allocate personnel, equipment, and other critical resources to areas at greatest risk, while improving evacuation planning during active fire events. Understanding the environmental factors most strongly associated with wildfire propagation can also help policymakers develop more effective mitigation strategies to strengthen long-term wildfire preparedness in vulnerable regions.

The input to our algorithms is a 64 × 64 × 12 float-valued tensor representing a 64 × 64 kilometer (km) geographic region at 1 km spatial resolution captured at a location and time when a wildfire occurred. The twelve channels consist of normalized difference vegetation index (NDVI), minimum temperature, maximum temperature, elevation, population density, wind speed, wind direction, specific humidity, precipitation, drought severity (Palmer Drought Severity Index), energy release component (ERC), and the previous-day wildfire mask. We use a baseline 1 × 1 CNN, a three-layer CNN, and a U-Net encoder-decoder CNN to predict the next-day wildfire mask, a 64 × 64 binary segmentation map indicating whether each pixel is predicted to contain wildfire 24 hours later. By comparing models with increasing spatial complexity, we evaluate how incorporating local and multi-scale spatial context improves wildfire spread prediction performance.

Related Work

Recent research has explored a variety of machine learning and deep learning approaches for predicting wildfire spread. Huot et al. (2022) developed the *Next Day Wildfire Spread* dataset, a curated, large-scale benchmark for predicting next-day wildfire spread using remote-sensing and environmental data collected across the United States. In addition to presenting the dataset, the authors established baseline performance using logistic regression, random forest, and CNN models. Their results demonstrated that the CNN outperformed the traditional machine learning models by leveraging the spatial relationships present in environmental variables and previous fire perimeters. A major strength of this work is the creation of a standardized, multivariate benchmark dataset that preserves the spatial structure of wildfire observations, providing a common benchmark for evaluating machine learning models. However, the benchmark models are limited to relatively simple CNN architectures and do not investigate deeper encoder-decoder networks capable of capturing multi-scale spatial context. Our work builds directly upon this dataset by comparing progressively more expressive CNN architectures, including a baseline 1 × 1 CNN, a three-layer CNN, and a U-Net encoder-decoder CNN.

A second line of research has explored encoder-decoder architectures for wildfire prediction. Singh et al. (2025) benchmarked an artificial neural network (ANN) against a U-Net architecture for wildfire susceptibility mapping, which estimates a

landscape's long-term propensity to burn, using geospatial variables such as elevation, vegetation indices, topography, and proximity to infrastructure. The authors found that the ANN achieved higher overall accuracy, while the U-Net captured spatial patterns more effectively, resulting in higher precision, recall, and F1 scores for identifying wildfire-prone locations. A key strength of this work is its demonstration that encoder-decoder architectures are particularly well suited for spatially detailed wildfire mapping tasks where minimizing false positives and false negatives is critical. However, the study focuses on long-term wildfire susceptibility within a single region of Australia rather than predicting the next-day spread of an active wildfire, making its findings less directly applicable to short-term wildfire propagation. Our work extends this line of research by evaluating a U-Net architecture on the nationwide *Next Day Wildfire Spread* dataset and comparing its performance against simpler CNN architectures.

Recent state-of-the-art wildfire prediction methods increasingly rely on deep learning architectures that learn complex nonlinear relationships while preserving the spatial structure of environmental and wildfire data. CNNs and encoder-decoder architectures are particularly effective because they learn relationships among neighboring pixels and across larger regions of an image, enabling them to capture both local and broader spatial context that influences wildfire spread. Building upon these approaches, our work evaluates three CNN architectures of increasing complexity on the *Next Day Wildfire Spread* dataset to determine how increasing spatial context influences next-day wildfire spread prediction performance.

Dataset

The Next Day Wildfire Spread dataset is a multivariate collection of historical wildfire data across the United States, distributed as TFRecord files. The dataset contains a total of 18,445 observations, pre-partitioned into training (14,979), validation (1,877), and test (1,689) sets. No missing values were found in any split. Each observation covers a 64x64 km region at 1km resolution, with 12 input features (NDVI, tmmn, tmmx, elevation, population, vs, pdsi, pr, sph, th, erc, PrevFireMask) and 1 target, FireMask (fire state 24 hours later; pixels are 1=fire, 0=no fire, -1=uncertain).

Data cleaning: Six features (tmmn, tmmx, th, vs, sph, erc) shared a common sentinel “no data” pattern—pixels set exactly to 0.0 across an unusually large, near-identical mask (~223K pixels). These were imputed with the training-set mean and applied consistently across splits to avoid data leakage. PDSI values outside its valid [-10,10] range were clipped; th and pr contained physically impossible values and were excluded from EDA rather than imputed. The 11 continuous features were z-scored using training set statistics; PrevFireMask was left unscaled as a discrete spatial mask. All EDA used training data only, with each feature collapsed to its spatial mean, plus two derived target variables: fire_pixel_fraction and has_fire.

Key Findings:

- Distributions (raw vs scaled values): Most features look like normal, moderately-skewed weather data. pr and population both show a huge spike at zero with a long thin tail, indicating that most locations receive no rain and have near-zero population. Elevation and erc are both bimodal; elevation spikes at 0 and around 2,000m (likely flat vs mountainous terrain) and erc spikes near 30 and 77 (indicating two different seasonal fire-weather sets). th’s raw values extend to -50,000 (invalid), so it was excluded. *(See Appendix for Graph A&B)*
- Box Plots (fire vs no fire): erc, elevation, psdi, tmmn, and tmmx show meaningful separation between classes. vs, sph, pr, and population show heavy overlap, which indicates a weak standalone correlation. CNN modeling might discover spatial patterns or combinations between features that box plots cannot. *(See Appendix for Graph C)*
- Correlation: erc (r = 0.41) is the strongest linear predictor, followed by elevation (0.23) and PrevFireMask (0.23), then pdsi (-0.21) and tmmx (0.20). PrevFireMask’s weaker-than-expected correlation suggests a nonlinear relationship, supporting a CNN/U-Net approach over linear models. *(See Appendix for Graph D)*
- Random Forest Feature Importance: elevation (0.30) and population (0.18) ranked highest despite weak correlations. Further checks showed substantial fire/no-fire distribution overlap (30-32%), indicating a class-imbalance distortion rather than a genuine relationship. Both features are kept for physical plausibility, not because of this ranking. *(See Appendix for Graph E)*
- Spatial Visualizations: fire spreads outward from PrevFireMask patches, and high erc regions coincide with fire growth, confirming local, spatial signal. "Uncertain" pixels sometimes border fire regions, suggesting they aren't randomly distributed (possibly cloud cover/sensor gaps). *(See Appendix for Graph F)*
- Class imbalance: Sample-level imbalance is moderate (88.6% fire / 11.4% no-fire), but pixel-level imbalance is severe (98.9% no-fire / 1.1% fire); median fire sample has fire in just 0.52% of pixels. Modeling used masked, weighted BCE (positive-class weight matched to the pixel-level ratio) and evaluated with AUC, precision, recall, F1, and IoU rather than accuracy.

Methods

All three of our algorithms utilize a CNN of various sizes to analyze the images in the dataset: the baseline model is a 1×1 CNN, the second model is a multilayer CNN, and the last model is a U-Net CNN. Each model uses the same hyperparameters: the

Adam optimizer, a learning rate of 0.001, and a batch size of 32, as well as the same binary cross-entropy (BCE) loss function. Due to the limited RAM of Google Colab, which we used to run our models, we used an adjusted loss function. We wrote a function, *masked_weighted_bce()*, that accomplishes two things while also computing BCE loss:

- The function ignores invalid pixels. These are pixels that are uncertain in our data and are not classified as either "fire" or "no fire."
- The function "upweights" the "fire" pixel class due to how rare it is. Because of this, the model won't simply learn to always predict "no fire."

Furthermore, each model uses the *EarlyStopping* checkpoint to monitor *val_loss* with a patience of 5 epochs and restores the best weights once the model has stopped running.

Additionally, we adjusted the training and validation datasets slightly to accommodate Google Colab's limited RAM. These datasets were created using a class we wrote, *ArraySequence*, which subclasses Keras's *Sequence* to create a custom generator that pulls batches directly from the existing in-memory *NumPy* arrays (X and y) via index slicing. This class also handles shuffling manually, meaning it reshuffles the indices of the training dataset after each epoch completes. Each epoch of training sees a completely different random ordering of batches, which helps prevent the model from partially memorizing batch-level patterns rather than generalizing.

Our baseline model is a 1×1 CNN, which is equivalent to a logistic regression model. This model applies a single *Conv2D* layer with a 1×1 kernel and a sigmoid activation directly to the 64×64×12 input, producing a 64×64×1 output of fire probabilities. Each output pixel is computed using only the twelve features values at that specific spatial location, and the model never looks at neighboring pixels for spatial context. This means the model learns a single set of twelve weights plus one bias term and applies the same linear combination and sigmoid formula independently to each pixel in the input grid. Each pixel is treated as one training example, with twelve input features, and the model learns the best linear decision boundary in that space with no spatial awareness. This model is a useful baseline for our problem because it establishes a floor for further improvement. Because our task is a spatial prediction problem, this baseline shows how much of the eventual U-Net encoder-decoder model performance comes from spatial reasoning over learning the right relationships between each pixel's features.

Our first improvement over the baseline is a multilayer CNN. This model stacks three *Conv2D* layers with 3×3 kernels, a ReLU activation, and filter sizes of 32 → 32 → 16. Each 3×3 convolution slides a learned filter across the input, calculating a combination of each pixel and its neighbors' feature values. After the first convolutional layer, each pixel's representation already involves spatial reasoning from its 3×3 neighborhood, so stacking two more convolutional layers allows for each output pixel's prediction to be influenced by a large surrounding neighborhood of size 5×5. The final layer outputs logits, rather than probabilities, to keep the output raw and unbounded. This model serves as a meaningful middle step between the per-pixel logistic regression baseline and the eventual U-Net architecture. It offers insight into whether or not adding local spatial context improves fire prediction compared to the baseline. Furthermore, because there is no pooling, downsampling, or upsampling, this multilayer CNN shows how much of the eventual U-Net's performance comes from its encoder-decoder architecture rather than local context alone.

Our second improvement over the baseline is a U-Net Encoder-Decoder CNN with skip connections. The encoder applies three stages of two-layer 3×3 convolutional blocks with filter sizes doubling from 16 → 32 → 64, which are then followed by a 2×2 max pooling layer after each stage, progressively shrinking the spatial resolution from 64×64 down to 8×8. The bottleneck reaches 128 filters, achieving a compressed representation of the input. The decoder then mirrors this process in reverse: it upsamples the feature map to a larger resolution, concatenates it with the corresponding encoder stage's saved output (the skip connection) to reinject the fine spatial detail the pooling removed, and passes it through another convolutional block. This process is repeated until the resolution returns to the initial 64×64 grid, at which point a final 1×1 convolution produces per-pixel logits. This model is appropriate for our problem because predicting wildfire spread is inherently a multi-scale spatial problem, and a U-Net architecture is designed to capture spatial context at multiple resolutions. Through the three stages of downsampling, the network builds abstract representations at each stage, providing the necessary context to "see" the larger surrounding regions. The decoder brings the broader context back up to full resolution using skip connections, which gets the model back to the needed per-pixel level.

Experiments, Results and Discussion

Setup

To compare the performance of our models fairly, we trained each architecture using the same training, validation, and test datasets. All models were implemented in TensorFlow using the Adam optimizer with an initial learning rate of 0.001 and a batch size of 32. Because the wildfire dataset contains invalid pixels labeled as -1 and a severe imbalance between wildfire and non-wildfire pixels, every model used the same custom masked weighted binary cross-entropy loss function. This allowed the models to ignore uncertain pixels during training while placing greater emphasis on wildfire pixels. To reduce overfitting, EarlyStopping was used to monitor validation loss and restore the best-performing model, while ModelCheckpoint saved the model with the lowest validation loss. Since

overall accuracy can be misleading for highly imbalanced datasets, we primarily evaluated performance using the Area Under the ROC Curve (also AUC). We also will report precision, recall, F1 score, and Intersection over Union (IoU) to better evaluate wildfire segmentation performance.

Baseline Model: 1×1 CNN

We first trained a 1×1 convolutional neural network to establish a baseline. Since this model makes predictions using only the environmental features at each individual pixel, it does not consider any neighboring information. This experiment allowed us to evaluate how much predictive power exists in the environmental variables alone before introducing spatial context. The baseline achieved a validation AUC of 0.7252. At the default classification threshold, the model obtained a precision of 0.0651, recall of 0.2495, an F1 score of 0.1032, and an IoU of 0.0544. While these results were better than random guessing, the model struggled to accurately identify wildfire pixels. This was expected because wildfire spread is inherently spatial, and treating each pixel independently prevents the model from learning how surrounding conditions influence fire movement.

Multi-Layer CNN

Our second experiment replaced the baseline with a deeper convolutional neural network consisting of three convolutional layers. Unlike the baseline, this model was able to incorporate information from neighboring pixels, allowing it to learn local spatial patterns associated with wildfire spread. Adding local spatial context produced a substantial improvement. The validation AUC increased from 0.7252 to 0.8211, showing that neighboring environmental conditions did contain valuable information for predicting wildfire spread. At the default threshold, the model achieved a precision of 0.0866, recall of 0.5741, an F1 score of 0.1504, and an IoU of 0.0813. Because of the class imbalance, we also evaluated different probability thresholds and selected the one that maximized the F1 score. Using the optimal threshold of 0.9024, precision increased to 0.2163, recall was 0.4073, the F1 score improved to 0.2825, and the IoU increased to 0.1645.

U-Net CNN

Our final architecture was a U-Net encoder-decoder network designed for image segmentation. Since wildfire spread depends on both local conditions and regional patterns, we expected the U-Net to outperform the previous models by learning features at multiple spatial scales. This architecture produced the strongest results of the three models. The validation AUC increased to 0.9303, representing a substantial improvement over the multi-layer CNN. Using the threshold that maximized the validation F1 score (0.9352), the model achieved a precision of 0.2695, recall of 0.4448, an F1 score of 0.3357, and an IoU of 0.2017. Compared to the previous models, the U-Net generated much stronger segmentation performance and was better able to distinguish wildfire pixels from the background.

Hyperparameter Tuning

After identifying the U-Net as the strongest architecture, we performed two additional tuning experiments. First, we added a SpatialDropout2D layer with a dropout rate of 0.15 to determine whether additional regularization would improve performance. Instead, the validation AUC decreased slightly from 0.9303 to 0.9239, so dropout was not included in the final model.
Next, we reduced the learning rate from 0.001 to 0.0003 while keeping all other hyperparameters unchanged. This produced the best-performing model in our project, increasing the validation AUC to 0.9341. At the optimal validation threshold (0.9103), the tuned model achieved a precision of 0.2715, recall of 0.4695, an F1 score of 0.3441, and an IoU of 0.2078. Although the improvement was smaller than the gains achieved by changing architectures, it showed that hyperparameter tuning could still improve model performance.

Final Model Evaluation

Finally, we evaluated the tuned U-Net on the held-out test dataset. The final model achieved a test AUC of 0.9365, which was slightly higher than the validation AUC of 0.9341. Because these values are very similar, the results tell us that the model generalized well to unseen data and showed little evidence of overfitting. On the test set, the model achieved a precision of 0.2761, recall of 0.5823, an F1 score of 0.3746, and an IoU of 0.2305. Across all of our experiments, we observed a progression in performance as additional spatial information was incorporated into the models. The baseline demonstrated that environmental variables alone contain useful predictive information, but treating each pixel independently limited performance. Adding convolutional layers allowed the model to learn local spatial relationships, producing a noticeable improvement. The U-Net architecture provided the largest gain by combining local and larger-scale spatial information, making it the best-performing model in our study. Finally, reducing the learning rate resulted in another small improvement, while adding dropout did not improve performance. Based on these results, the tuned U-Net with a learning rate of 0.0003 was selected as our final model because it consistently achieved the strongest performance on both the validation and test datasets.

Conclusions

In this project, we developed and compared three convolutional neural network models to predict next-day wildfire spread using environmental, weather, topographic, and previous-day wildfire data. Overall, our results showed that including spatial information made a big difference in prediction performance. The baseline 1×1 CNN was able to learn some useful patterns from the environmental variables, but it struggled because it treated every pixel independently. The multi-layer CNN improved on these results by learning relationships between neighboring pixels, while the U-Net performed the best by capturing both local and larger-scale spatial features through its encoder-decoder architecture. After tuning the learning rate, the final U-Net achieved the strongest performance with a test AUC of 0.9365, an F1 score of 0.3746, and an IoU of 0.2305. The similar validation and test results also suggest that the model generalized well and did not show much evidence of overfitting.

There are still several limitations to this project. The dataset was super imbalanced, with wildfire pixels making up only a small portion of the data, which made the segmentation task much more difficult. We were also limited by the computational resources available through Google Colab, so we couldn't explore larger models or a wider range of hyperparameters. If we had more time and computing power, we think it would be interesting to test deeper U-Net architectures, try different and new loss functions such as Dice Loss or Focal Loss, and perform a more extensive hyperparameter search. It would also be worthwhile to incorporate additional temporal information or newer attention-based models to better capture how wildfires spread over time. The project showed that deep learning can be an effective approach for short-term wildfire prediction, and there is still plenty of room to improve the model in future work.

Contributions

Aida Mohasesi:

Developed the Wildfire_Prediction_Aida.ipynb notebook. Parsed the data into NumPy arrays. Conducted data preprocessing and EDA. Developed a baseline 1x1 CNN model and a U-Net model. Wrote a rough draft of each section of the Project Milestone. Combined all versions of the methodology section written by each team member into one section for the Project Milestone. Wrote the Abstract, Introduction, and Related Work sections of the Project Report and helped finalize all other sections of the Project Report.

Jordan Agresti:

Developed the Wildfire_Prediction_Jordan.ipynb notebook. Conducted data preprocessing and EDA. Developed a baseline 1x1 CNN, multi-layered CNN, and U-Net CNN model. Wrote a rough draft of each section of the Project Milestone, and combined all team members' versions of "Data Preprocessing" into one section for the Project Milestone. Wrote the Dataset section of the Project Report and helped finalize all other sections of the Project Report

Aria Koul:

Developed the WildfirePrediction_Aria_eda.ipynb, WildfirePrediction_Aria_preprocessing.ipynb, and WildfirePrediction_FinalModel.ipynb notebooks. Parsed the data into NumPy arrays and conducted preprocessing and EDA. Developed a baseline 1x1 CNN, a multilayer CNN, and a U-Net CNN. Evaluated each model and performed hyperparameter tuning. Wrote a rough draft of each section of the Project Milestone, and combined all team members' versions of the "Data Challenges" into one section for the Project Milestone. Wrote the Methods section of the Project Report and helped finalize other sections.

Molly Jenkins:

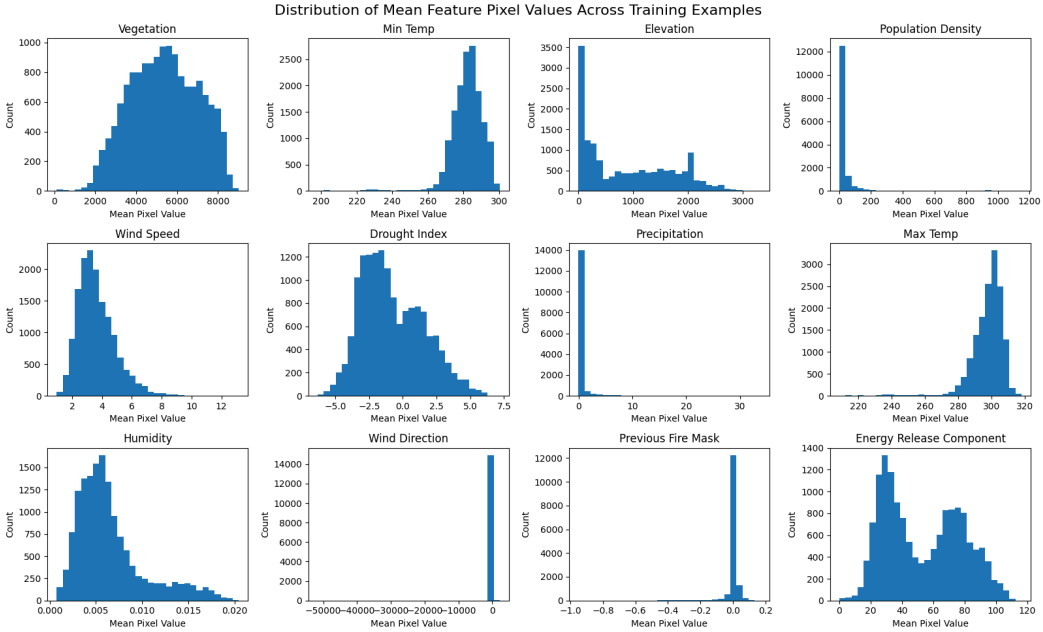
Developed WildfirePrediction_MollyJenkins.ipynb. Conducted EDA and data pre-processing. Wrote a rough draft for the project milestone and combined all group members work into one section for the EDA section of the final report. Wrote Experiments, Results and Discussion and Conclusion sections of the final report.

References

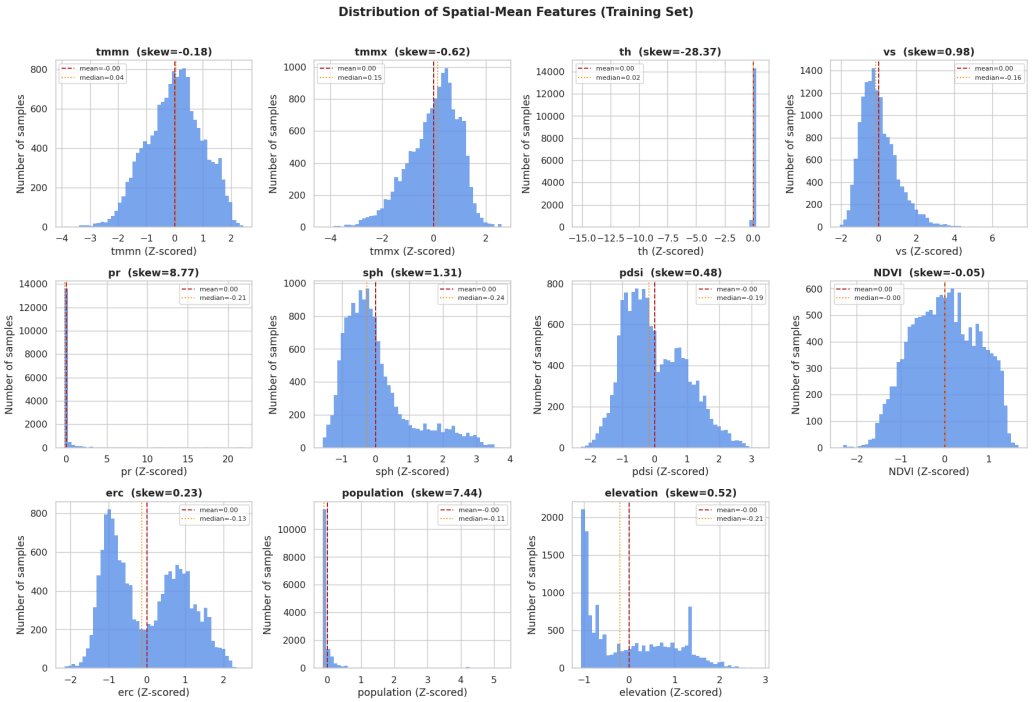
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- [2] Singh, H., Ang, L. M., & Srivastava, S. K. (2025). Benchmarking artificial neural networks and U-net convolutional architectures for wildfire susceptibility prediction: Innovations in geospatial intelligence. *IEEE Transactions on Geoscience and Remote Sensing*, *63*, 1-15.

Appendix

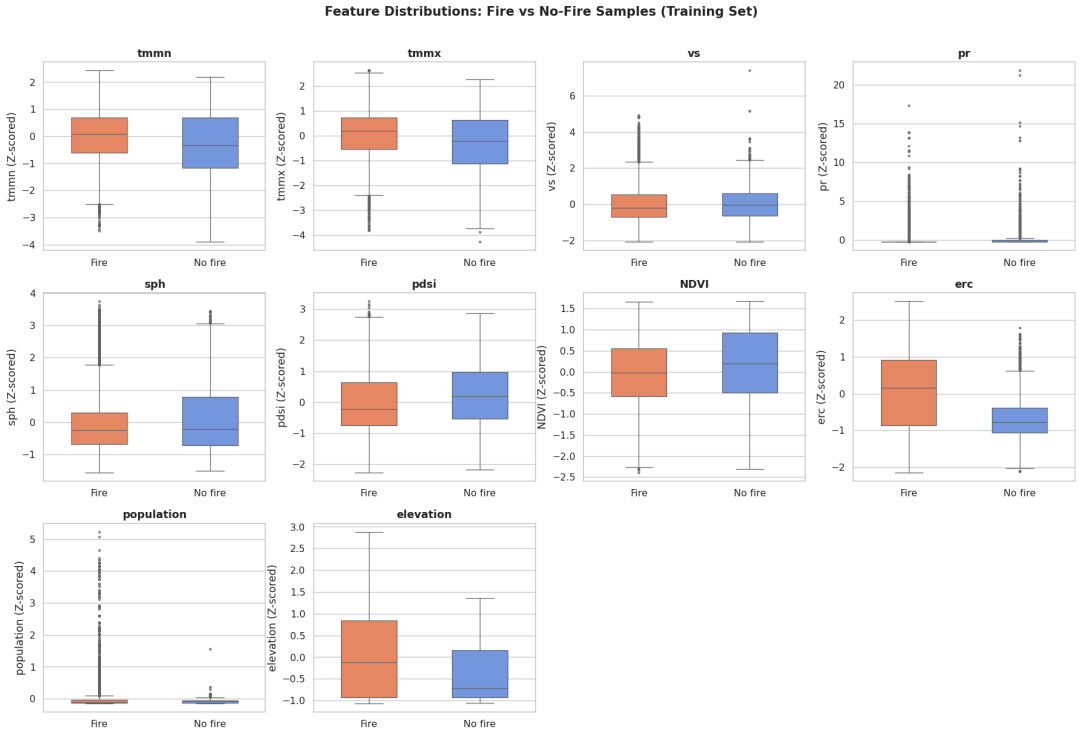
Graph A: Distribution of Mean Feature Pixel Values (Raw Values)



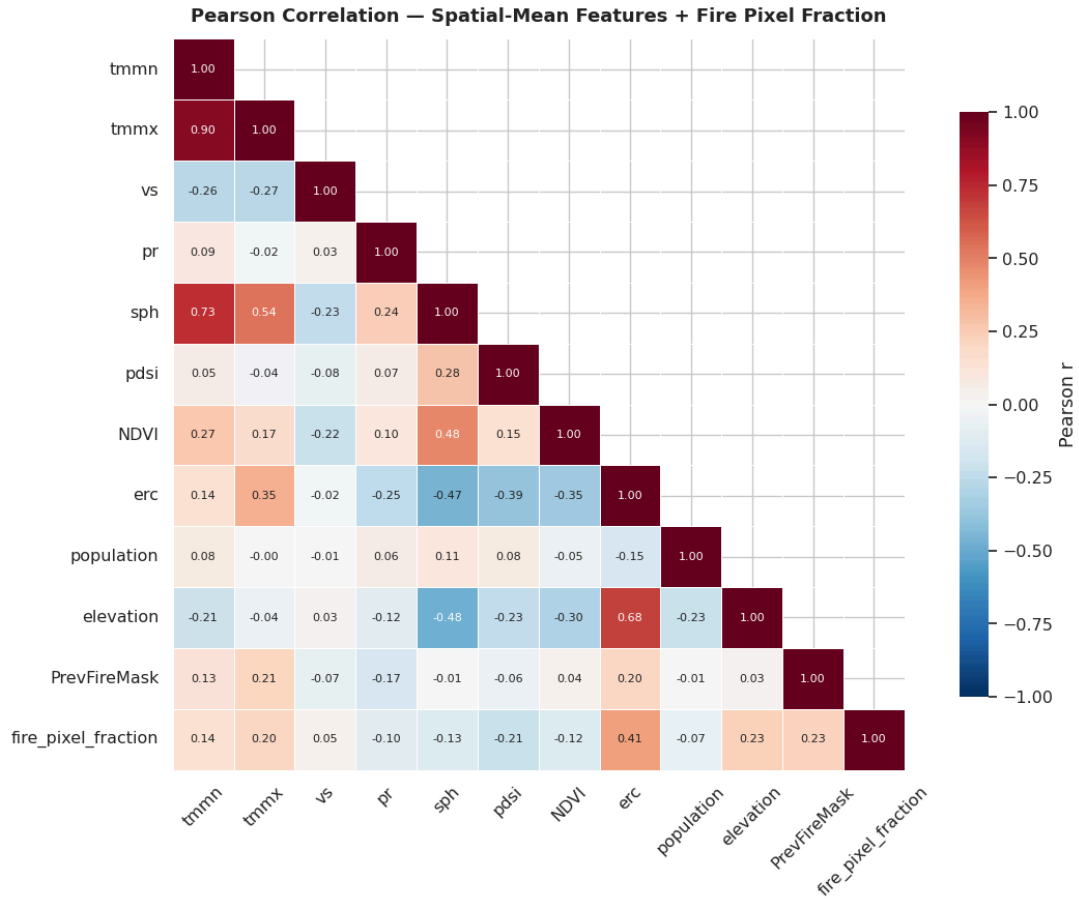
Graph B: Distribution of Mean Feature Pixel Values (Standardized/Z-Scored)



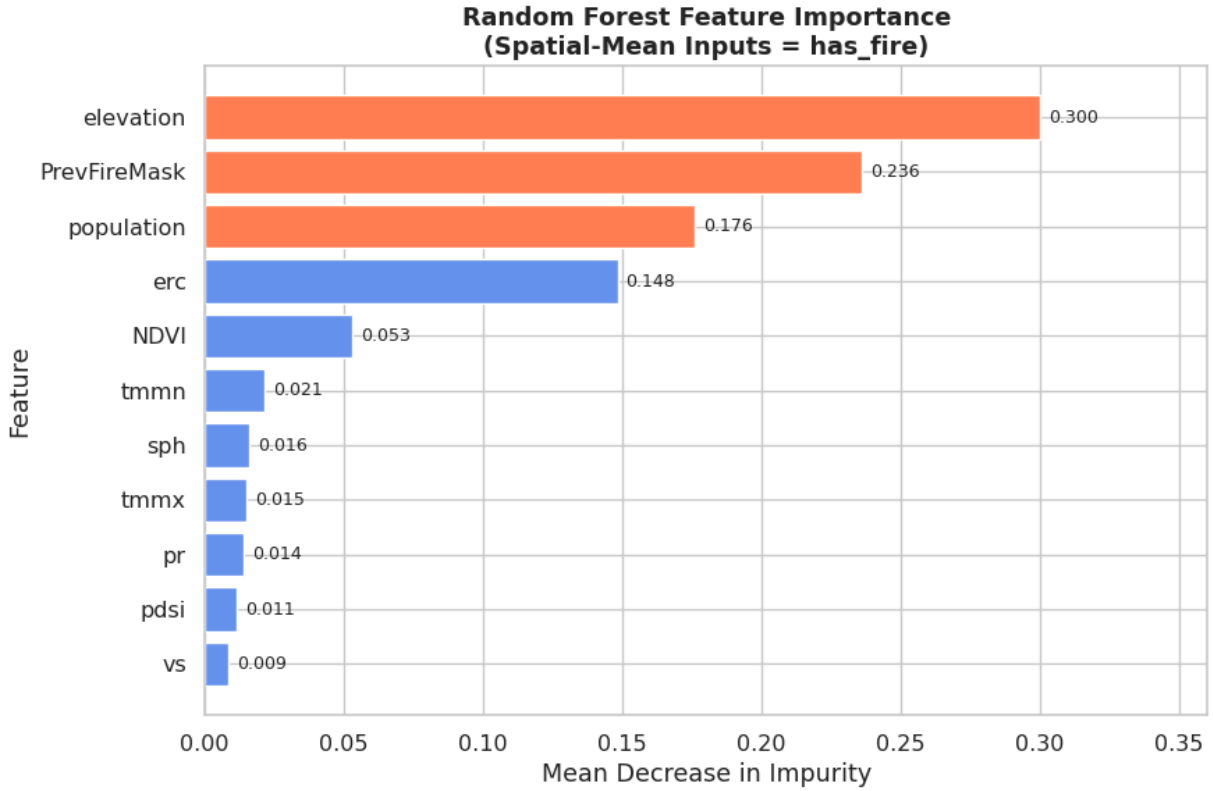
Graph C: Box Plot (Fire vs No-Fire Feature Distributions)



Graph D: Correlation Heatmap (Pearson)

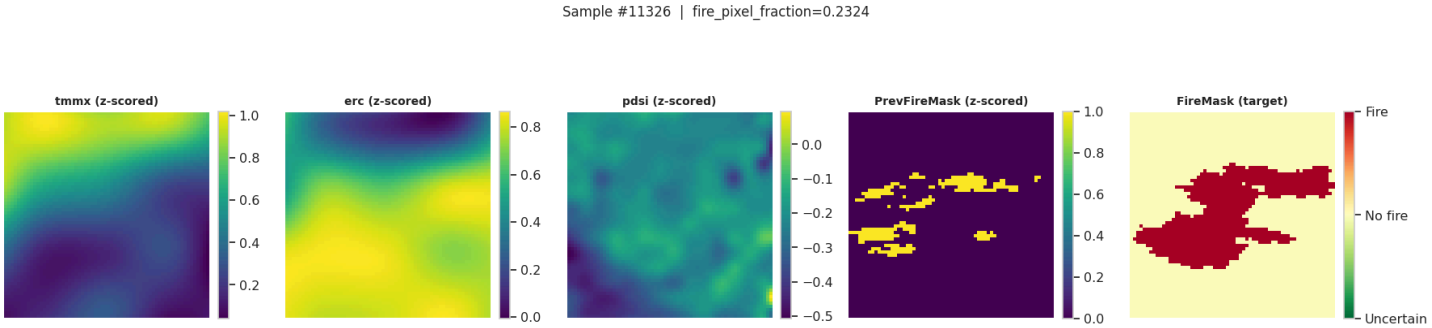


Graph E: Random Forest Feature Importance

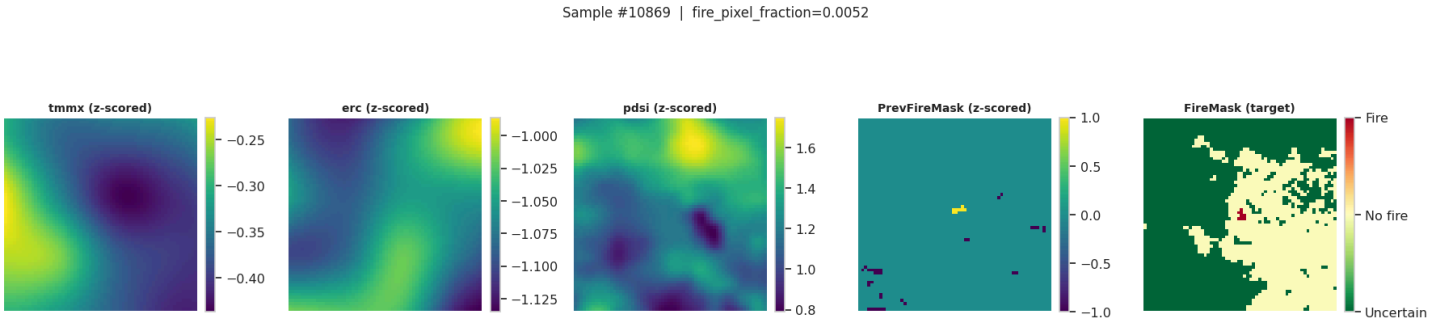


Graph F: Spatial Visualizations

Largest Fire:



Medium Fire:



Smallest Fire:

